

Auteur: Seda Jusjajeva
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$$\begin{aligned} & \int \frac{1}{\sqrt{x^2 - 4}} dx \\ &= \int \frac{1}{\sqrt{4 \left(\frac{x^2}{4} - 1 \right)}} dx \\ &= \frac{1}{2} \int \frac{1}{\sqrt{\frac{x^2}{4} - 1}} dx \\ &= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{x}{2} \right)^2 - 1}} dx \end{aligned}$$

Substitutie: $t = \frac{x}{2} \Rightarrow dt = \frac{1}{2} dx$

$$\begin{aligned} &= \int \frac{1}{\sqrt{t^2 - 1}} dt \\ &= \ln |t + \sqrt{t^2 - 1}| + c \\ &= \ln \left| \frac{x}{2} + \sqrt{\frac{x^2}{4} - 1} \right| + c \end{aligned}$$

References